

QSVM Track: Week 5 Progress Summary

Four-model baselines, subsample validation, kernel diagnostics,
and the first significance tests

2 August 2026

Abstract

This is the Week 5 report for the QSVM track. Three of its results revise conclusions this track had previously drawn rather than merely extending them. First, the classical baseline is no longer SVM alone: on EMBER binary a random forest beats SVM at every $n_{\text{components}}$, so the quantum–classical margins reported up to now were *understated*. Second, the collapse of the QSVM to chance performance on the CIC 15-class task, previously recorded as unexplained, is attributed here to *kernel concentration* and demonstrated against an RBF control on identical features. Third, all cross-framework claims are now supported by paired significance tests, which had been specified in Week 1 and never run: all 84 testable comparisons favour the classical model at $p < 0.05$. We also close the subsample-representativeness validation that every quantum result in this project had silently assumed for four weeks.

1 Data and protocol

Two malware datasets, each carrying a binary and a 15-class task.

CIC-MalMem-2022

Obfuscated-malware memory dumps, 55 numeric forensic features (process, handle, callback and DLL counts). The binary task uses all 58 596 rows and is exactly balanced between benign and malware. The 15-class task drops benign rows — detecting benign is the binary task’s job — leaving 29 298 malware rows across 15 families, mildly imbalanced at $1.7\times$ between the largest and smallest.

EMBER 2018

Static PE features, 2381 dimensions. Work is done on a 18 014-row extract of the published test set; the full set is a single $200\,000 \times 2384$ block that cannot be read incrementally. The binary task uses all 18 014 rows and is 88.9% malware. The 15-class task keeps families with at least 500 samples, caps at the 15 largest, and downsamples each to the smallest kept count, giving 14 325 rows exactly balanced at 955 per family.

Shared evaluation. The quantum kernel costs $O(n^2)$ circuit evaluations to build, so every quantum result is computed on a stratified 1000-row subsample rather than the full data. The subsample and the five stratified outer folds computed over it are written to disk by the quantum runner and *replayed* by the classical runner, so both frameworks are scored on identical rows and identical folds — 800 training and 200 held-out rows per fold. §4 verifies that this subsample is faithful to its population.

Both frameworks also consume one shared feature pipeline: a zero-variance filter, a correlation filter at 0.95, standardisation, then PCA to $n_{\text{components}}$ components. That single parameter fixes the classical feature count and the quantum qubit budget simultaneously, and is what makes the comparison like-for-like. The pipeline is fit on training-fold rows only.

Metric. All results are held-out macro-F1, the unweighted mean of per-class F1, reported as a mean over the five outer folds. Accuracy is not used as a headline: on EMBER binary at $n_{\text{components}} = 1$ the QSVM attains 0.890 accuracy in three of five folds while the benign class’s F1 is exactly 0.000 in those same folds, because 88.9% of rows are malware and the model labels everything malware. Accuracy there reports near-success for a classifier that never once identifies the class of interest.

2 What this week changes

Earlier work in this track compared the QSVM against SVM alone, left the CIC 15-class collapse mechanism unexplained, and reported no p -values. Table 1 summarises what has changed.

Table 1: Items this track had left open, and their status now.

Position entering this week	Now	Evidence
Classical comparison is SVM only	Resolved	Four models at $n_{\text{components}} \in \{1, 3, 6\}$, both tasks (§3)
CIC 15-class collapse mechanism unidentified	Attributed	Kernel concentration, against an RBF control (§5)
No significance testing	Resolved	84 paired tests (§6)
Subsample fidelity assumed	Verified	KS tests far below chance (§4)
Duplicated family-selection logic	Resolved	Single shared function
SOREL blocked	Closed	Resume conditions recorded (§7)
Test suite: 119 tests	144 tests	—

A metric label to correct. Figures reported for the EMBER binary task up to now have been labelled “accuracy”; they are macro-F1. The distinction is the one drawn in §1: the two differ by more than 0.4 on that task, and only one of them reflects whether the benign class is ever found.

3 Four-model classical baseline on EMBER

Every EMBER quantum–classical margin on record was a QSVM-versus-SVM margin. Adding random forest, XGBoost and LightGBM at $n_{\text{components}} \in \{1, 3, 6\}$ closes that gap. All runs replay the persisted quantum subsample and folds, so the four models and the QSVM are scored on identical rows.

Table 2: EMBER 2018, held-out macro-F1 (mean $\pm$ std over 5 outer folds). Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_{\text{components}} = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_{\text{components}} = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_{\text{components}} = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_{\text{components}} = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_{\text{components}} = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_{\text{components}} = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

The previously reported binary margins were understated. On the binary task the random forest beats SVM at every $n_{\text{components}}$, so comparing the QSVM against SVM measured it against a sub-optimal baseline:

	$n_{\text{components}} = 1$	$n_{\text{components}} = 3$	$n_{\text{components}} = 6$
gap vs. SVM (as previously reported)	+0.0727	+0.1222	+0.1587
gap vs. best classical	+0.0995	+0.1491	+0.1697
understated by	0.0268	0.0269	0.0110

On the 15-class task SVM genuinely is the strongest baseline at $n_{\text{components}} \in \{3, 6\}$, so those two margins stand unchanged. Four of six cells move, all in the same direction: every corrected cell moves against the quantum model on EMBER.

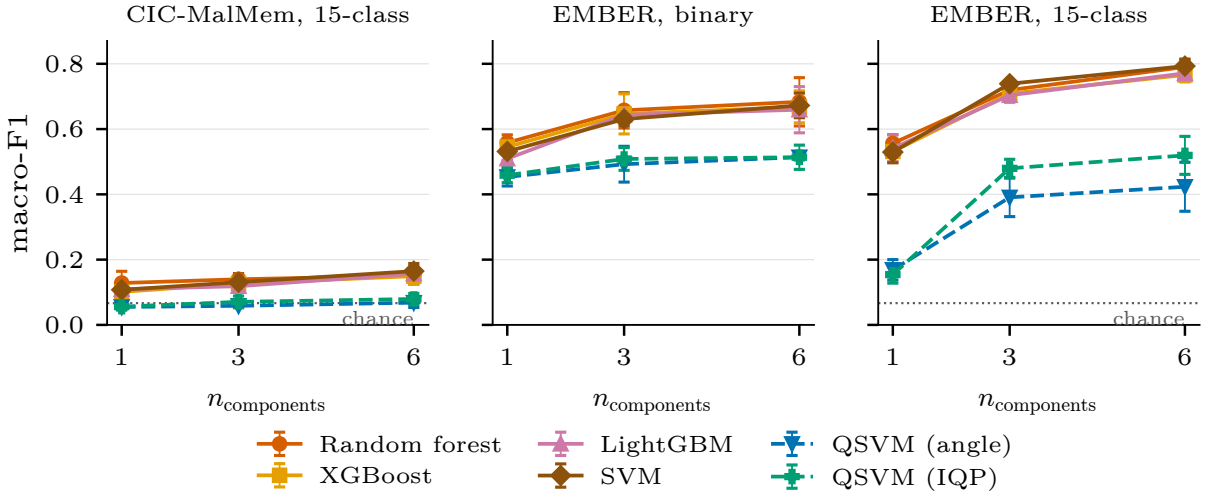


Figure 1: Held-out macro-F1 against the shared feature/qubit budget. Solid lines are classical models, dashed lines the QSVM encodings; error bars are one standard deviation across the five outer folds. CIC binary is absent because no sweep of it can be grouped (§6). Note that the classical-quantum gap on EMBER binary *widens* with $n_{\text{components}}$: the QSVM gains 0.055 macro-F1 from one to six components where the random forest gains 0.125.

4 Subsample representativeness

Every quantum result in this project is computed on a 1000-row stratified subsample, forced by the $O(n^2)$ kernel cost. That the subsample represents its population was specified as a Week 2 task, never executed, and assumed for four weeks. It is now measured.

The decisive statistic is the two-sample Kolmogorov–Smirnov rejection count read *against its own null expectation*, since a perfectly faithful subsample still produces about $\alpha \cdot n_{\text{features}}$ rejections by chance. A bare count is uninterpretable; the ratio is not.

Both subsamples are faithful. Four weeks of quantum results now rest on a measured foundation rather than an assumption.

5 Why CIC 15-class collapses

That summary established the contrast — the QSVM sits at the random baseline on CIC 15-class while reaching 0.42–0.52 on EMBER 15-class — and ruled out class imbalance and qubit count, closing with “the actual cause is unidentified”. Two measurements now separate a dataset effect from a kernel effect.

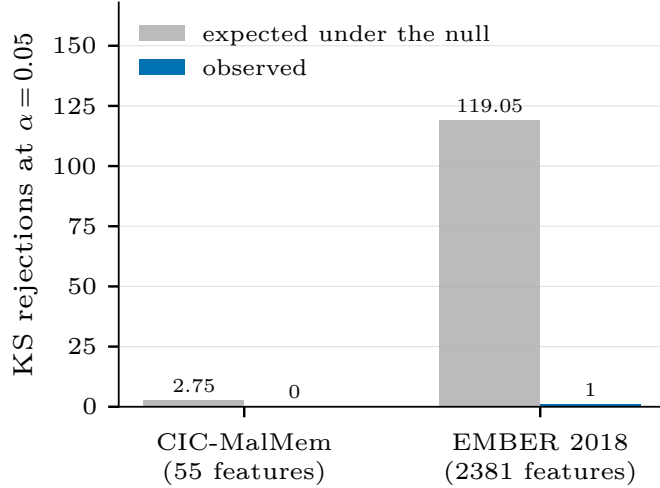


Figure 2: Per-feature KS rejections at $\alpha = 0.05$, subsample against full population, beside what chance alone produces. CIC rejects nothing where ≈ 2.75 were expected; EMBER rejects one feature where ≈ 119 were expected. Maximum class-proportion drift is below 0.001 on both.

5.1 Feature geometry explains the dataset, not the kernel

Measured on the identical projected features both frameworks consume, fit on training folds only: EMBER’s families are $\approx 8\times$ more separable by one-vs-rest Fisher ratio, and up to $50\times$ on the worst-case family. CIC’s silhouette coefficient is *negative at every* $n_{\text{components}}$: the average CIC point lies closer to another family’s centroid than to its own.

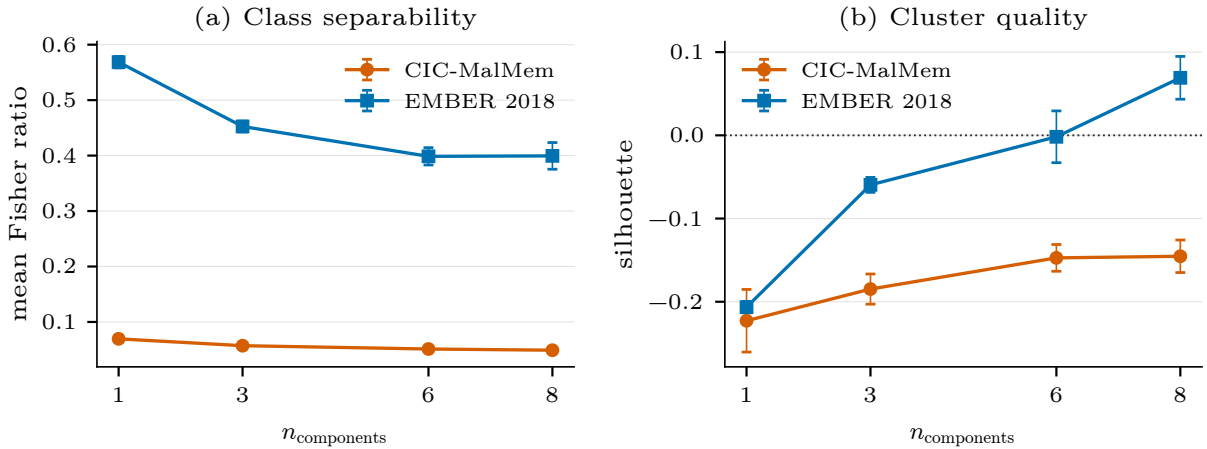


Figure 3: Class separability of the shared PCA projection, averaged over the five outer folds. Both metrics are scale-invariant and therefore comparable across datasets of very different dimensionality.

Counter-intuitively, CIC’s PCA explains substantially *more* variance (33–81%) than EMBER’s (8–30%) while separating families far less — CIC’s variance is concentrated in directions that do not discriminate family, which is plausible for a memory-forensics feature vector whose process and handle counts track workload rather than malware family.

This explains why *every* method does badly on CIC 15-class. It cannot explain the QSVM-specific failure, because a classical SVM extracts 0.108–0.165 macro-F1 from exactly these features where the QSVM extracts 0.056–0.079.

5.2 Kernel geometry explains the kernel

The decisive experiment builds a classical RBF Gram matrix on the *identical* feature matrix, which separates “these features are hard” from “this kernel is bad”. Both diagnostics are computed on the first outer training fold at $n_{\text{components}} = 6$ with the IQP encoding (800 rows, 6 qubits).

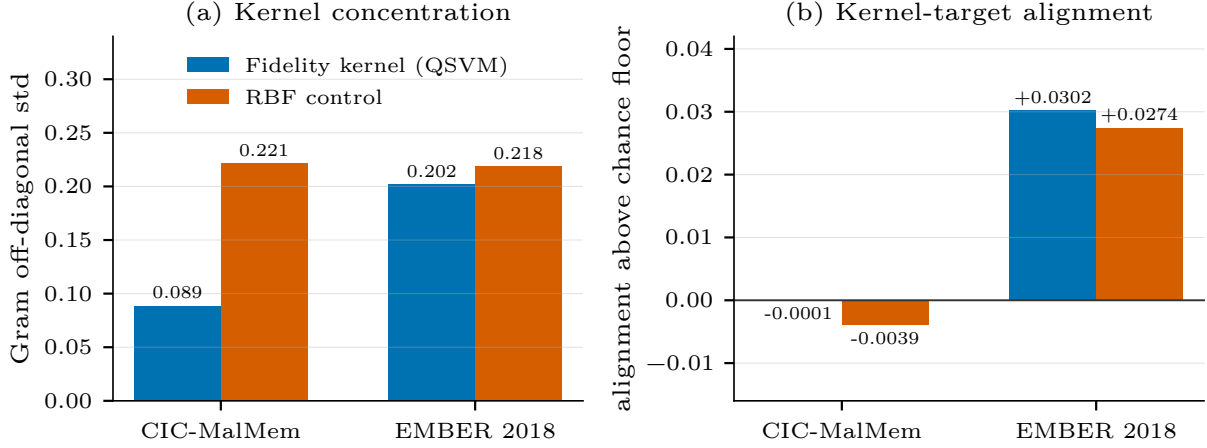


Figure 4: (a) Standard deviation of the off-diagonal Gram entries: as this collapses, the Gram approaches a scaled identity and the SVM has nothing to separate on. (b) Kernel-target alignment relative to its floor. A constant kernel scores $1/\sqrt{15} \approx 0.258$ on 15 balanced classes, so the raw alignment is nearly uninformative and the excess is what carries meaning.

Alignment separates the datasets, not the kernels. On CIC the fidelity kernel lands 0.000069 *below* the constant-kernel floor — indistinguishable from an all-ones matrix — and the RBF control is below it too. On EMBER both clear it by ≈ 0.03 .

Concentration separates the kernels, specifically on CIC. The RBF kernel produces near-identical spread on both datasets (0.2212 against 0.2184, a 1% difference), so CIC’s features do not make a classical kernel concentrate. The fidelity kernel tracks RBF on EMBER ($0.92\times$) but collapses to $0.40\times$ its spread on CIC. Every off-diagonal similarity is squeezed into a band $2.5\times$ narrower from the same inputs.

An unresolved tension. Alignment reports that neither kernel’s Gram carries label information on CIC, yet a classical SVM reaches 0.16 there. Kernel-target alignment is a global, unweighted average over all pairs, and an SVM requires no such thing — it requires support vectors, a regularisation constant and class weights, none of which alignment models. Alignment is a useful measure of relative dataset difficulty and an unreliable predictor of achievable SVM performance. The concentration result does not share this weakness, because it describes the Gram the SVM actually receives.

What is and is not supported. Supported: the fidelity kernel’s failure on CIC is a property of *the kernel*, not of the dataset. Not supported: that tuning **bandwidth** would repair it. Bandwidth is the identified mechanism and has never been tuned on CIC beyond its default, but no sweep was run. This is a hypothesis with a named lever, not a measured fix.

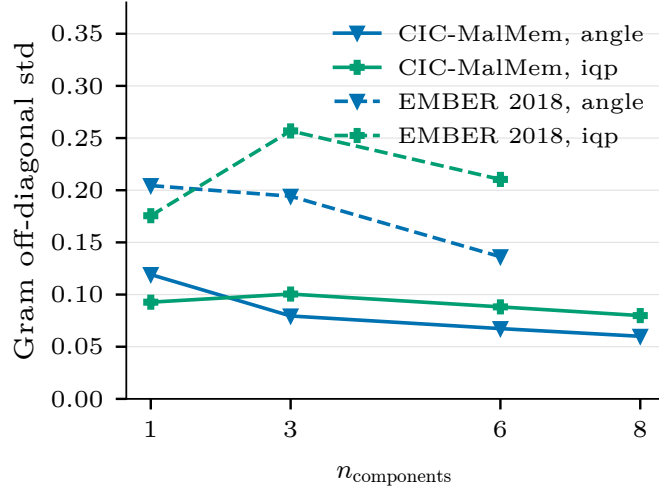


Figure 5: Concentration across the full 15-class sweep. This required no new computation: the estimator has logged the off-diagonal spread on every fit since Week 2. EMBER’s Gram is $1.9\text{--}2.6\times$ more spread than CIC’s at every comparable setting, and concentration worsens with qubit count on both — the trend the `bandwidth` parameter exists to counteract.

6 Paired significance tests

Significance testing was specified in Week 1, built and unit-tested, and never run. Four weeks of “QSVM loses to classical” rested on differences in mean macro-F1 with no test that any gap exceeded fold-to-fold noise.

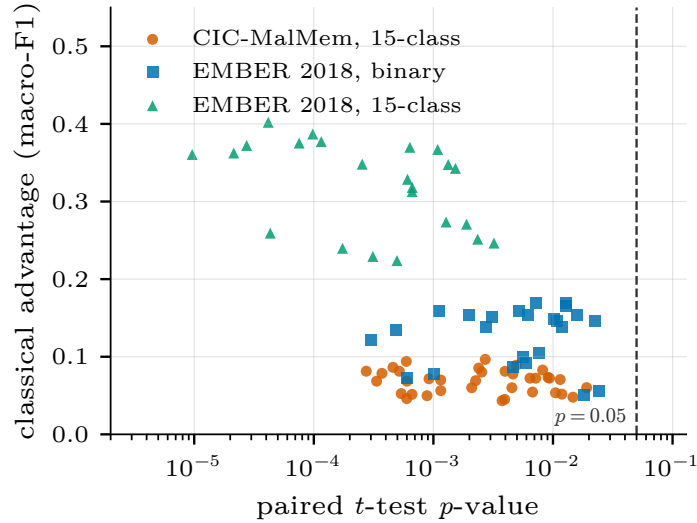


Figure 6: Effect size against paired t -test p -value for all 84 testable quantum–classical pairs. Every point lies left of $p = 0.05$ and above zero: all 84 favour the classical model at conventional significance. The worst p -value across every pair is 0.0241.

6.1 Wilcoxon returned 0.0625 for all 84 pairs, and that is not a null result

With five paired samples, the Wilcoxon signed-rank test’s smallest attainable two-sided p -value is 0.0625. It is therefore *mathematically incapable* of reaching $p < 0.05$ at this fold count, regardless of effect size. All 84 pairs returning exactly 0.0625 means every pair achieved the most extreme rank configuration available — all five folds favouring the classical model. Reporting “Wilcoxon found nothing significant” would be a serious misreading. Raising the fold count to six would

put 0.03125 within reach; that, not a different test, is the remedy.

6.2 What could not be tested

- **All CIC binary comparisons** (9 cells $\times$ 4 models). Every CIC binary run — classical *and* quantum, 45 fold records per model — was logged before the experiment tracker recorded parent/child run structure, and carries no recoverable sweep boundary. The raw records show the QSVM at 0.92–1.00 macro-F1, consistent with the earlier “ties on binary” reading, but they resolve into two distinct sweeps per $n_{\text{components}}$ with different configurations and no identifier separating them. Grouping them by time-gap clustering would invent a boundary the data does not record; no attempt was made. The limitation is symmetric across frameworks, not a quantum-specific gap.
- **McNemar between frameworks.** The test needs per-sample predictions from both models on a shared test set. QSVM predictions were not persisted during these sweeps. Persistence has since been added, but no sweep was re-run, so nothing is recoverable retroactively. Classical-versus-classical McNemar was run: on EMBER 15-class the random forest beats XGBoost at $p = 0.0022$, while on CIC 15-class *no pair separates at all* — consistent with §5.

6.3 A methodological note on selecting comparable runs

Identifying which logged records constitute one cross-validation run is not mechanical, and getting it wrong silently corrupts every aggregate. Filtering on (dataset, task, $n_{\text{components}}$, model, encoding) — the obvious approach — fails twice over:

1. The same configuration has legitimately been swept several times over the project’s weeks, so a parameter-based grouping returns 10, 15 or 20 records for a five-fold sweep and averages distinct hyperparameter searches together. Of the parameter groups in the experiment export, 57 have a size other than five.
2. For the QSVM, *encoding is a per-fold outcome, not a configuration key*: the two-tier tuner selects the winning encoding inside each fold, so a single joint sweep can appear as “four angle folds and one IQP fold”. Grouping on encoding splits one sweep into two incomplete ones.

Runs are therefore grouped by parent run, keeping only sweeps whose parent completed with exactly five folds; where several clean sweeps match, the most recent wins. Under this rule the store holds 125 nested sweeps, 123 of them clean. The two exceptions — an interrupted sweep and a truncated three-fold sweep from a past out-of-memory incident — are excluded automatically rather than deleted, leaving the history intact.

7 SOREL-20M closed

SOREL-20M is closed for this track rather than deferred. Its multiclass labelling scheme was designed and validated over 19.7M rows and remains delivered work; the feature store was never fetched and *no SOREL sweep was ever run*. The two are independent, and the labelling report should not be read as partial results.

Measured at close-out: 227 GiB of free disk against a 71.6 GiB requirement, so disk is not the blocker. Memory is not either — the store is a memory-mapped LMDB whose reads page through the operating system cache, which the earlier phrasing (“the whole file must be local before any row can be read”) is easy to misread as a memory constraint. It is a locality constraint. The blocker is the fixed 71.6 GiB transfer, unavoidable because LMDB offers no key-level remote access: there is no way to fetch only the thousand rows a sweep would need.

8 Summary

The project’s central claim is stronger than it was and points the same way. Classical models beat the QSVM on every testable configuration, now with $p < 0.05$ on all 84 of them rather than on mean differences alone, and the EMBER margins turn out to have been understated rather than overstated. The one configuration where the quantum pipeline was ever competitive — CIC binary — cannot be significance-tested from what was logged.

What is genuinely new is the mechanism. The fidelity kernel does not fail on CIC because CIC’s features are hard: a classical RBF kernel on the identical features does not concentrate. It fails because the fidelity kernel itself collapses to $0.40\times$ the classical kernel’s spread on that data. That is a statement about the method rather than about the dataset, and it is the most transferable result this track has produced.

Reproduction. Every figure is generated programmatically from the same recorded experiment metrics the tables are drawn from, so no figure can disagree with a reported number. The implementation carries 144 passing tests.